Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Primary, secondary, tertiary amines can be separated by the following except:

(A) Fractional distillation

(B) Fractional method using diethyl oxalate

(C) Hinsberg's method using $C_6H_5SO_2Cl$

(D) Selective crystallisation

Q.2

Hinsberg reagent is

(A) $C_6H_5SO_8Cl$

(B) $C_6H_5SO_4Cl$

(C) $C_6H_5SO_2Cl$

(D) $C_6H_5SO_3Cl$

Q.3

Total number of reagents among the following which can reduce $CH_3CH_2NO_2$ in to ethyl amine.
$Fe/HCl, Sn/HCl, H_2/Ni, NH_2NH_2/H_3O^{\oplus}, B_2H_6/THF,$
$LiAlH_4, NaBH_4, Zn/Hg + HCl$

(A) 1

(B) 5

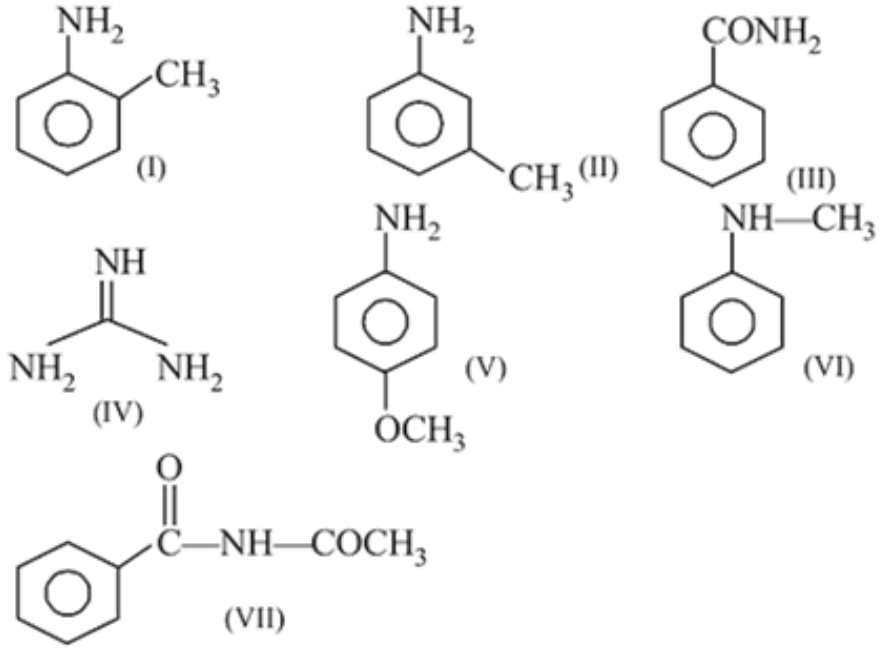
(C) 2

(D) 3

Q.4

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Total number of compounds which are more basic than aniline.



(A) 2

(B) 5

(C) 4

(D) 6

Q.5

The conversion of benzene diazonium chloride to bromobenzene can be accomplished by

(A) Reimer-Tiemann reaction

(B) Friedel-Crafts reaction

(C) Gattermann reaction

(D) Azo-Coupling reaction

Q.6

C_7H_9N has how many isomeric forms that contain a benzene ring?

(A) 4

(B) 5

(C) 6

(D) 7

Q.7

The amine that does not react with acetyl chloride is

(A) CH_3NH_2

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(B) $(CH_3)_2NH$

(C) $(CH_3)_3\ N$

(D) None of these

Q.8

In the diazotisation of arylamines with sodium nitrite and hydrochloric acid, an excess of hydrochloric acid is used primarily to

(A) supress the concentration of free aniline available for coupling

(B) supress hydrolysis of phenol

(C) ensure a stoichiometric amount of nitrous acid

(D) neutralise the base liberated

Q.9

Nitrosoamines $(R_2N-N=O)$ are insoluble in water. On heating with conc. H_2SO_4, they give secondary amines. The reaction is called

(A) Liebermann nitroso reaction

(B) Etard reaction

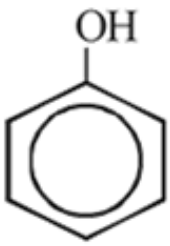
(C) Fries reaction

(D) Perkin reaction

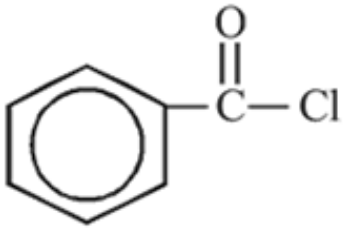
Q.10

Aniline reacts with phosgene and KOH to form

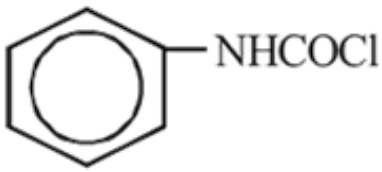
(A)



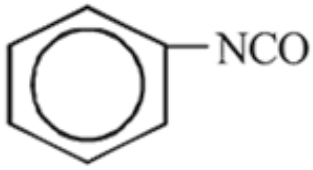
(B)



(C)

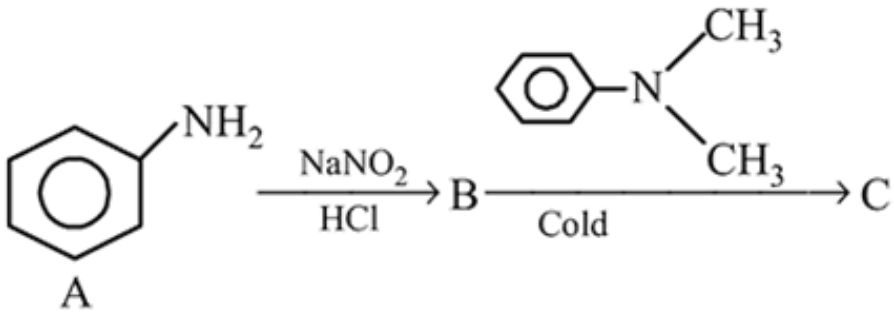


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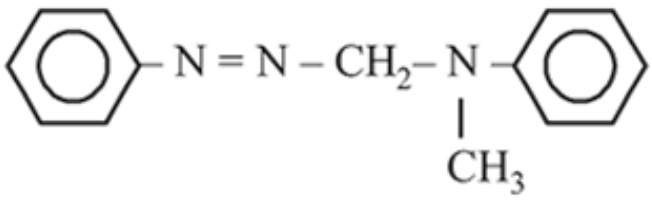
(D) 

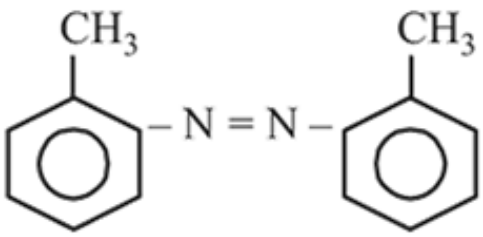
Q.11

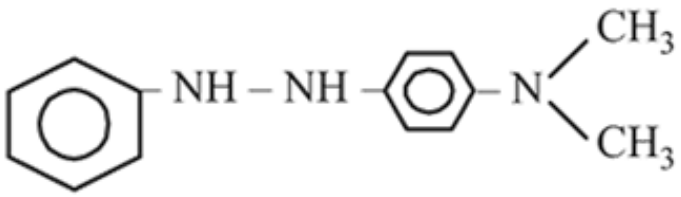
In a reaction of aniline a coloured product C was obtained.

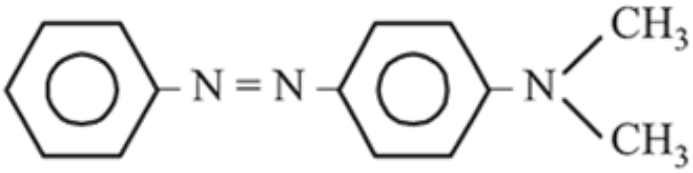


The structure of C would be:

(A) 

(B) 

(C) 

(D) 

Q.12

Reaction of cyclohexanone with dimethylamine in the presence of catalytic amount of an acid forms a compound if water during the reaction is continuously removed. The compound formed is generally known as

(A) an amine

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(B) an imine

(C) an enamine

(D) a Schiff's base

Q.13

p -Chloroaniline and anilinium hydrogen chloride can be distinguished by:

(A) Sandmeyer reaction

(B) Carbylamine reaction

(C) Hinsberg's reaction

(D) $AgNO_3$

Q.14

Primary amines can be distinguished from secondary and tertiary amines by reacting with

(A) Chloroform and alcoholic KOH

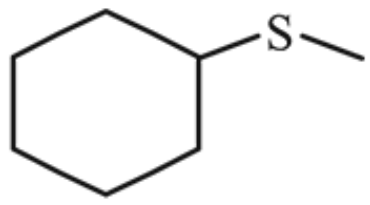
(B) Methyl iodide

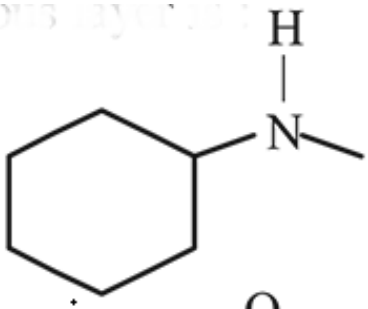
(C) Chloroform alone

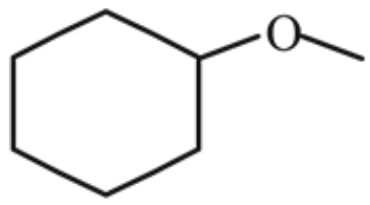
(D) Zinc dust

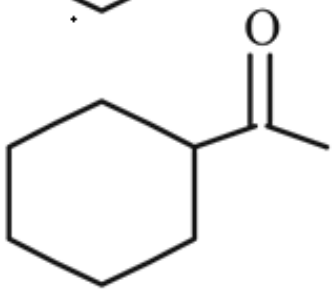
Q.15

A mixture containing the following four compounds is extracted with $1MHCl$. The compound that goes to aqueous layer is :

(I) 

(II) 

(III) 

(IV) 

(A) (I)

(B) (II)

(C) (III)

(D) (IV)

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Q.1 Answer:

Selective crystallisation

Solution:

Separation of $1^{\circ}, 2^{\circ}$ and 3° amines:

$1^{\circ}, 2^{\circ}$ and 3° amines $+\mathrm{R_4NX} \xrightarrow{\text{distillation}}$ Mixture

Mixture of $1^{\circ}, 2^{\circ}, 3^{\circ}$ amine can be separated by following methods.

(i) Fractional distillation: The mixture of amines may be fractional distillation because their boiling points are quite different. It is used in industry.

(ii) Hinsberg method: In this method mixture of amines is seperated by using benzene sulphonyl chloride (Hinsberg's reagent).

$\mathrm{C_6H_5SO_2Cl}$ **+ 1° amine $\rightarrow$ Product** $\xrightarrow{\mathrm{KOH}}$ **dissolve**

$\mathrm{C_6H_5SO_2Cl}$ **+ 2° amine $\rightarrow$ Product** $\xrightarrow{\mathrm{KOH}}$ **insoluble**

3° amine does not react with benzene sulphonyl chloride.

(iii) Hofman method: In this method mixture of amines is separated by using ethyl oxalate.

1° amine + ethyl oxalate $\rightarrow$ solid product

2° amine + ethyl oxalate $\rightarrow$ liquid product

3° amine + ethyl oxalate $\rightarrow$ No reaction

Q.2 Answer:

$\mathrm{C_6H_5SO_2Cl}$

Solution:

$\mathrm{C_6H_5SO_2Cl}$

Q.3 Answer:

5

Solution:

$\mathrm{Fe/HCl, Sn/HCl, H_2/Ni, LiAlH_4}$ **and** $\mathrm{Zn - Hg + HCl}$

Q.4 Answer:

4

Solution:

II, IV, V and VI are more basic than aniline.

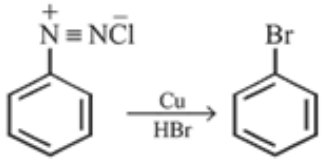
Q.5 Answer:

Gattermann reaction

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Solution:

Diazonium salts react with copper powder and hydrogen bromide to form bromobenzene. This reaction is known as Gattermann reaction.

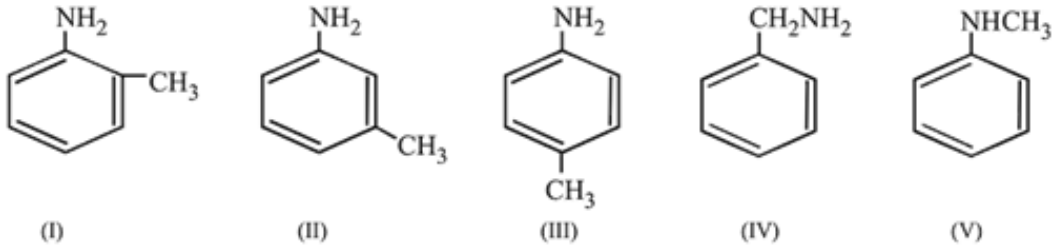


Q.6 Answer:

5

Solution:

$C_7H_9\ N$ **having one benzene ring may be represented as** $C_6H_5CH_4\ N$**, may be in the form of** 1° **and** 2° **amines in the following five isomeric forms.**



(I) (II) (III) (IV) (V)

Q.7 Answer:

$(CH_3)_3\ N$

Solution:

The compounds containing active H **-atoms** $(H$ **atoms attached to** N, O **or** $S)$ **react with** CH_3COCl **to form acetyl derivatives.**

Q.8 Answer:

supress the concentration of free aniline available for coupling

Solution:

Excess of HCl **is used to convert free aniline to aniline hydrochloride otherwise free aniline would undergo coupling reaction with benzenediazonium chloride.**

Q.9 Answer:

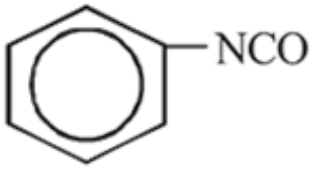
Liebermann nitroso reaction

Solution:

$$R_2NNO + H_2O \rightarrow R_2NH + HNO_2$$

Q.10 Answer:

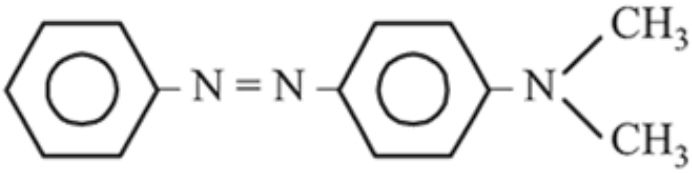
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Solution:

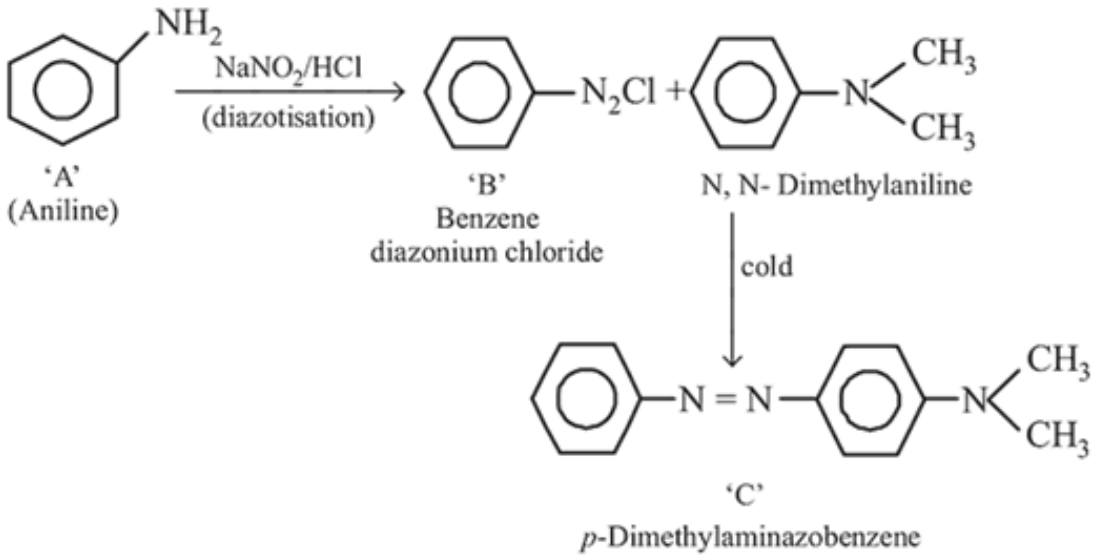
$$C_6H_5NH_2 + COCl_2 + KOH \rightarrow C_6H_5NH \cdot COCl + HCl \xrightarrow{\Delta} C_6H_5NCO + HCl$$

Q.11 Answer:



Solution:

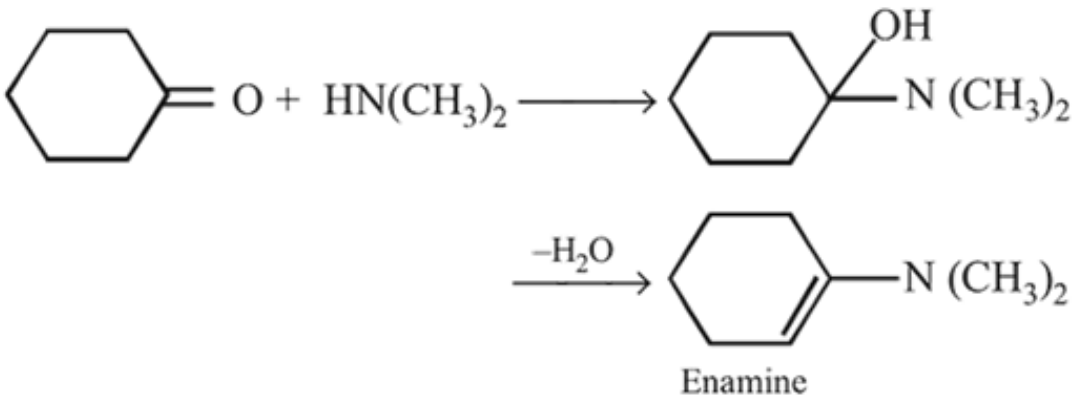
The reaction can be completed as follows:



Q.12 Answer:

an enamine

Solution:



Q.13 Answer:

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$AgNO_3$

Solution:

> ***p*-Chloroaniline and anilinium hydrogen chloride can be distinguished by $AgNO_3$. Anilinium hydrogen chloride will give white ppt of $AgCl$ on reaction with $AgNO_3$, but *p*-chloronoaniline will not react with it because Cl is directly attached to benzene nucleus.**

Q.14 Answer:

Chloroform and alcoholic KOH

Solution:

> **Primary amine reacts with $CHCl_3$ and alc. KOH to form isocyanide while secondary and tertiary amines do not react**

Q.15 Answer:

(II)

Solution:

> **When the given mixture is shaken with $1M$ HCl, amine get protonated and becomes cation $\left(RNH_2^{\oplus}\right)$, which does not dissolve in organic solvent but usually dissolve in H_2O due to its charge. So, shaking with aqueous HCl will pull amines into the aqueous phase and leave all other compounds in organic layer.**

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